JONAS EMILIO PETERSEN

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Head of Research at Forgis, a $5M pre-seed startup building the industrial world model. Built a 10-person team with 10 papers accepted at NeurIPS and ICML 2026 (1 oral, 1 spotlight), a $300k compute grant and a place in the Google DeepMind Accelerator. Declined a BCG offer to co-found K2 AI (>€500k revenue in 10 months). Cambridge MPhil (Distinction), Imperial MEng (Dean's List).

EXPERIENCE

Forgis — Zurich, Switzerland
Head of Research (since May 2026); Founding ML Engineer, employee #1 — *Oct 2025 – Present*

- Defined the research direction with the CTO; partnering with 5 of the world's top-10 universities
- Built and lead a 10-person research team (5 MSc, 2 PhD, 1 postdoc, 2 professors)
- Won a $300k CSCS compute grant (30,000 GPU hours on Alps) and the Google DeepMind Robotics Accelerator
- Deployed Hyperion with Gemma 3n on one edge GPU, featured as a Google DeepMind Gemmaverse case study
- 3rd of 5,000+ teams and 1st in Europe in the Intrinsic AI for Industry Challenge ($20k prize)

ETH Zürich — Zurich, Switzerland
Researcher, Autonomous Systems Lab (Prof. Siegwart), part-time — *Feb 2026 – Present*

- Funded by the Gebert Rüf Stiftung to accelerate Forgis' industrial world model research

K2 AI — London, UK
Co-founder — *Sep 2023 – Dec 2024*

- Co-founded an AI consultancy for media companies; grew to 10 people and >€500k revenue in 10 months
- Conceptualised 4 AI products with feedback from 30+ CFOs; built a task-mining SaaS MVP on SLMs
- Reintegrated part of the team into the parent organisation

Boston Consulting Group — Munich, Germany
Visiting Associate — *Jun – Aug 2021*

- Built a PowerBI cost-forecasting tool; received a full-time offer, declined in 2023 to co-found K2

SELECTED PUBLICATIONS

- **FactoryBench**: Evaluating Industrial Machine Understanding — arXiv
  Y. Merzouki, C. Izquierdo, . . . , **J. Petersen**†, P. Petersen† | *Oral*, PUDM @ NeurIPS 2026
- **Saturn**: A Time-Series Foundation Model for Classification and Event Forecasting
  J. Petersen, et al. | FMTS @ NeurIPS 2026
- **KITE**: A Knowledge-Injected Temporal Encoder for Time-Series Event Forecasting — arXiv
  J. Petersen, P. Jawahar, et al. | FMTS @ NeurIPS 2026
- **Certified Access**: Test-Time Compute on Conservative Physics-Structured AI Memory — arXiv
  P. Jawahar, R. Maggioni, **J. Petersen**, et al. | LCFM @ NeurIPS 2026
- **Retention You Can Predict, Scope and Delete** — OpenReview
  P. Jawahar, R. Maggioni, **J. Petersen**, et al. | STODY @ NeurIPS 2026
- **HEPA**: A Self-Supervised Horizon-Conditioned Event Predictive Architecture for Time Series — arXiv
  J. Petersen, G.-A. Lombardi, et al. | *Spotlight*, FMSD @ ICML 2026
- **FactoryNet**: A Large-Scale Dataset toward Industrial Time-Series Foundation Models — arXiv
  K. Othman, **J. Petersen**, et al. | AI4Physics and FMSD @ ICML 2026
- **Tabula RASA**: What Structural Inductive Bias Helps Transformers Reason Over Knowledge Graphs? — arXiv
  J. Petersen, C. Mazzoleni, et al. | GFM @ ICML 2026
- **Modelling and Optimisation of Extinction Actions for Wildfire Suppression** (MEng thesis) — PDF
  J. Petersen, S. Kapur, S. Gkantonas, E. Mastorakos, A. Giusti | *Combustion Science & Technology* (2023)

† Shared last authorship. Team total: 10 papers accepted at NeurIPS and ICML 2026.

EDUCATION

University of Cambridge — Cambridge, UK
MPhil, Nanotechnology – Distinction (highest honours) — *Sep 2022 – Jul 2023*

Imperial College London — London, UK
MEng, Mechanical Engineering – Distinction, Dean's List (top 10%) — *Sep 2018 – Jul 2022*

Sachsenwaldschule Reinbek — Hamburg, Germany
Abitur 1.0 (96.33%), valedictorian; best student in Maths, Chemistry and Physics — *Jul 2017*

SKILLS & INTERESTS

ML & robotics: PyTorch, Python, JEPA, self-supervised learning, time-series foundation models, LLMs, RL, ROS
Other: German (native), English (fluent); tennis, chess, jazz; robot learning (2 hackathon wins, YC interview)